

$$T(n) = T(n-1) + n^2 \rightarrow O(n^3)$$

we are multiply by n

$$T(n) = T(n-2) + 1 \rightarrow \frac{n}{2} \rightarrow O(n)$$

$$T(n) = T(n-100) + 1 \rightarrow \frac{n}{100} \Rightarrow O(n^2)$$

$$T(n) = 2T(n-1) + 1 \rightarrow ?$$

Ex: 2.2

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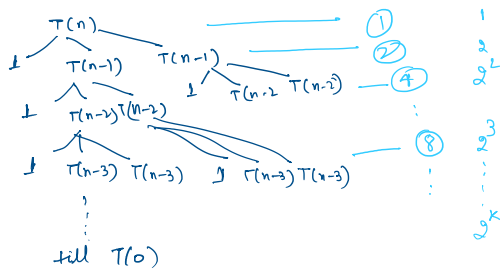
Test(n)  — T(n)
{
  if(n > 0)
  {
    print() — 1
    Test(n-1) — T(n-1)
    Test(n-1) — T(n-1)
  }
}

```

$$T(n) = 2T(n-1) + 1$$

recurrence relation

$$T(n) = \begin{cases} 1 & n=0 \\ 2T(n-1) + 1 & n>0 \end{cases}$$



$$= 1 + 2 + 2^2 + 2^3 + \dots + 2^k$$

$$= 2^{k+1} - 1$$

GP series $a + ar + ar^2 + ar^3 + \dots + ar^k = \frac{a(r^{k+1} - 1)}{r - 1}$

$$a = 1 \quad r = 2$$

$$\frac{1(2^{k+1} - 1)}{2 - 1} = 2^{k+1} - 1$$

assume $n = k = 0$

$$= 2^{n+1} - 1$$

$$O(2^n) \text{ or } O(2^n)$$

using substitution

$$T(n) = 2 \times T(n-1) + 1 \quad \text{--- (1)}$$

$$T(n) = 2 \times [2T(n-2) + 1] + 1$$

$$T(n) = 2^2 T(n-2) + 2 + 1 \quad \text{--- (2)}$$

$$T(n) = 2^2 [2T(n-3) + 1] + 2 + 1$$

$$= 2^3 T(n-3) + 2^2 + 2 + 1 \quad \text{--- (3)}$$

⋮ k times

$$T(n) = 2^k T(n-k) + 2^{k-1} + 2^{k-2} + \dots + 2 + 1 \quad (4)$$

Assume $n-k=0$
 $n=k$

$$= 2^n T(0) + 2^{n-1} + 2^{n-2} + \dots + 2 + 1$$

$$= 2^n + 2^{n-1} + 2^{n-2} + \dots + 2 + 1$$

$$= 2^{n+1} - 1$$

$$O(2^n) \text{ or } \Theta(2^n)$$

Master Theorem for Decreasing Functions.

$$T(n) = T(n-1) + 2 = O(n)$$

$$T(n) = T(n-2) + n = O(n^2)$$

$$T(n) = T(n-1) + \log n = O(n \log n)$$

$$T(n) = 2T(n-1) + 1 = O(2^n)$$

$$T(n) = 3T(n-1) + 1 = O(3^n)$$

$$T(n) = 2T(n-1) + 1 = O(n 2^n)$$

$$T(n) = aT(n/b) + f(n)$$

$a > 0$ $b > 0$ and $f(n) = O(n^k)$ where $k > 0$

$$\textcircled{1} \text{ if } a=1 \quad O(n^{k+1}) \text{ or } O(n \cdot f(n))$$

$$\textcircled{2} \text{ if } a > 1 \quad O(n^k a^{n/b})$$

$$\textcircled{3} \text{ if } a < 1 \quad O(n^k) \text{ or } O(f(n))$$

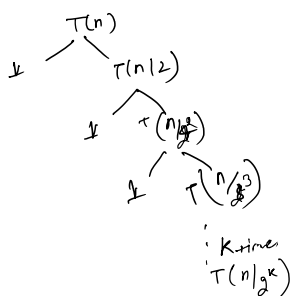
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Dividing functions

$$\begin{array}{l} T_{\text{est}}(n) \quad \text{---} \quad T(n) \\ \downarrow \\ \text{if } (n > 1) \\ \downarrow \\ \text{Pivot} \\ T_{\text{est}}(n/2) \quad \text{---} \quad T(n/2) \\ \downarrow \\ \downarrow \\ \hline T(n) = T(n/2) + 1 \end{array}$$

$$T(n) = \begin{cases} 1 & n=0 \\ T(n/2) + 1 & n > 1 \end{cases}$$

Solve Tree method



We stopped when $\frac{n}{2^k} = 1$

$$n = 2^k$$

we have taken k steps.

$$\log n = k$$

Substitution Method

$$T(n) = T(n/2) + 1 \quad (1)$$

$$= \left[T\left(\frac{n}{2^2}\right) + 1 \right] + 1 \quad T(n/2) = T(n/2) + 1$$

$$= T\left(\frac{n}{2^2}\right) + 2 \quad (2)$$

$$T\left(\frac{n}{2^2}\right) = T\left(\frac{n}{2^2}\right) + 1$$

$$= T\left(\frac{n}{2^3}\right) + 3 \quad (3)$$

$$= T\left(\frac{n}{2^4}\right) + 4 \quad (4)$$

$$T(n) = T\left(\frac{n}{2^k}\right) + k \quad (5)$$

assume $\frac{n}{2^k} = 1$ as we terminated

$$n = 2^k$$

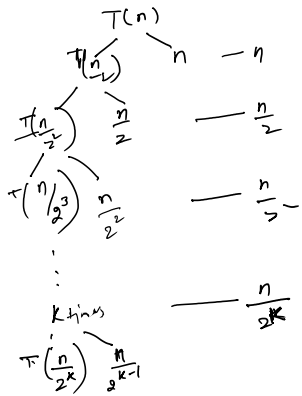
$$k = \log n$$

$$T(n) = T(1) + \log n$$

$$T(n) = 1 + \log n$$

$$O(\log n)$$

$$\text{ep} \quad \tau(n) = \begin{cases} 1 & n=1 \\ \tau(n/2) + n & n>1 \end{cases}$$



$$T(n) = n \left(1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^k} \right)$$

$$= n \left(\sum_{i=0}^k \frac{1}{2^i} \right) \text{ is } 2$$

$$= n \times 1$$

$$= n$$

$$\tau(n) = \tau(n/2) + n$$

$$\tau(n) = \left[\tau\left(\frac{n}{2^k}\right) + \frac{n}{2} \right] + n$$

$$= \tau\left(\frac{n}{2^k}\right) + n + \frac{n}{2}$$

$$= \left[\tau\left(\frac{n}{2^k}\right) + \frac{n}{2^k} \right] + n + \frac{n}{2}$$

$$= \tau\left(\frac{n}{2^k}\right) + n + \frac{n}{2} + \frac{n}{4} + \dots + \frac{n}{2^k}$$

$$\therefore k + m \text{ is } 0$$

$$= \tau\left(\frac{n}{2^k}\right) + \frac{n}{2^k} + \frac{n}{2^{k-1}} + \dots + \frac{n}{2^2} + \frac{n}{2} + n$$

$$\text{Assume } \frac{n}{2^k} = 1$$

$$n = 2^k$$

$$\log n = k$$

$$\tau(n) = \tau(1) + n \left[\frac{1}{2^{k-1}} + \frac{1}{2^{k-2}} + \dots + \frac{1}{2} + 1 \right]$$

$$= 1 + n(1+1)$$

$$= 1 + 2n$$

$$O(n)$$

$$\text{ep3} \quad \text{Test}(n) \quad \text{---} \quad T(n)$$

```

if (n > 1)
{
    for (i = 0; i < n; i++)
    {
        sum += i;
    }
    Test(n/2);
    Test(n/2);
}

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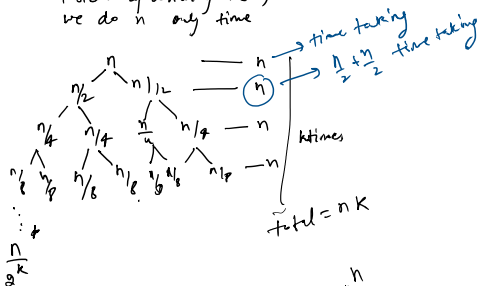
$$T(n) = n + 2(T(n/2))$$

$$T(n) = \begin{cases} 1 & n=1 \\ n + 2T(n/2) & n>1 \end{cases}$$

$$T(n) \rightarrow n$$

$$T(n/2) \rightarrow n/2$$

instead of writing $T(n)$
we do n only time



assume $\frac{n}{2^k} = 1$

$$n = 2^k$$

$$\log n = k$$

$$n \log n$$

$$O(n \log n)$$

$$T(n) = 2T(n/2) + n \quad \text{--- (1)}$$

$$= 2 \left[2T\left(\frac{n}{2^2}\right) + \frac{n}{2} \right] + n$$

$$= 2^2 T\left(\frac{n}{2^2}\right) + n + n \quad \text{--- (2)}$$

$$= 2^2 \left[2T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} \right] + n + n$$

$$= 2^3 T\left(\frac{n}{2^3}\right) + n + n + n \quad \text{--- (3)}$$

$$= 2^k T\left(\frac{n}{2^k}\right) + n \cdot k \quad \text{--- (4)}$$

assume $\frac{n}{2^k} = 1$

$$n = 2^k$$

$$\log n = k$$